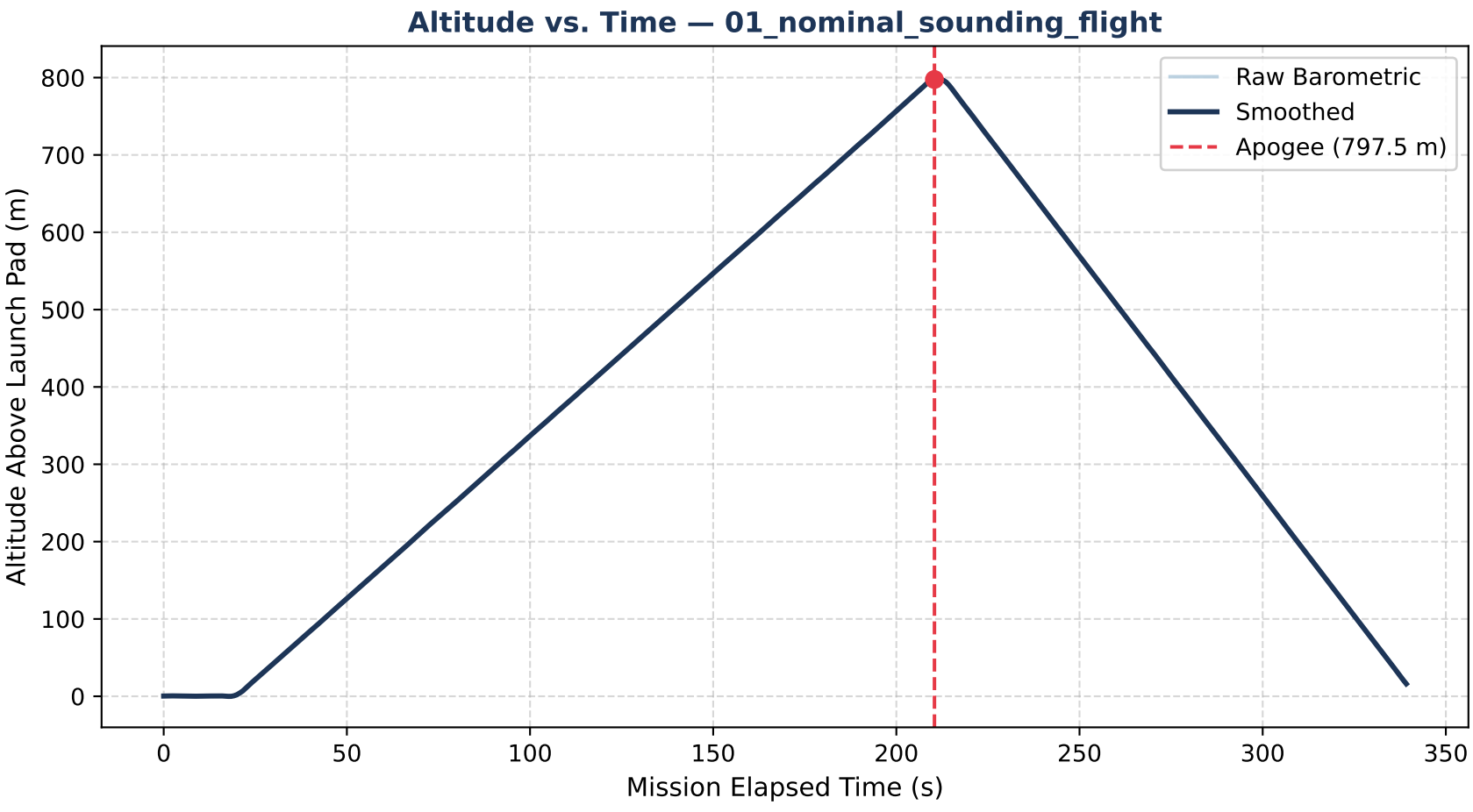
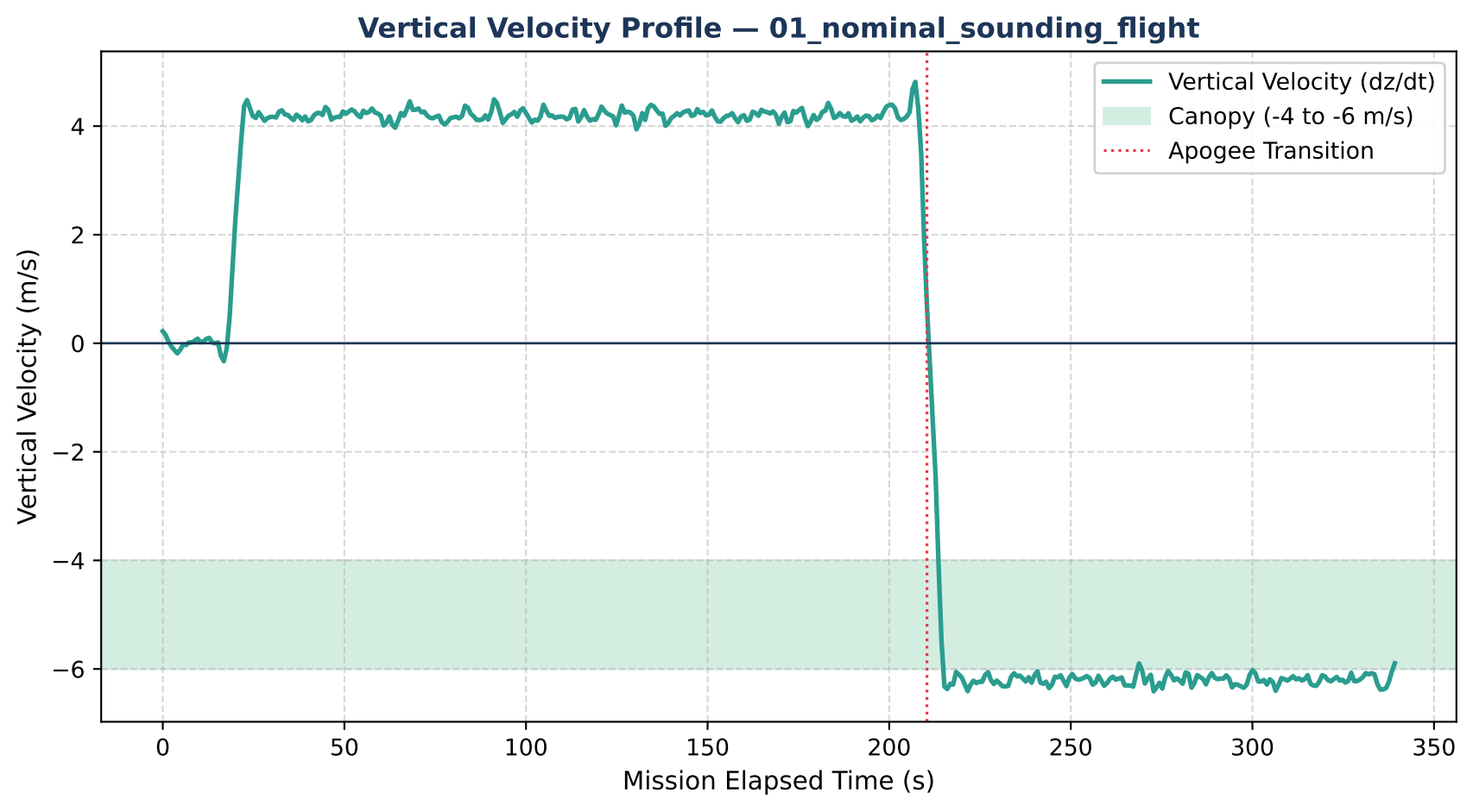
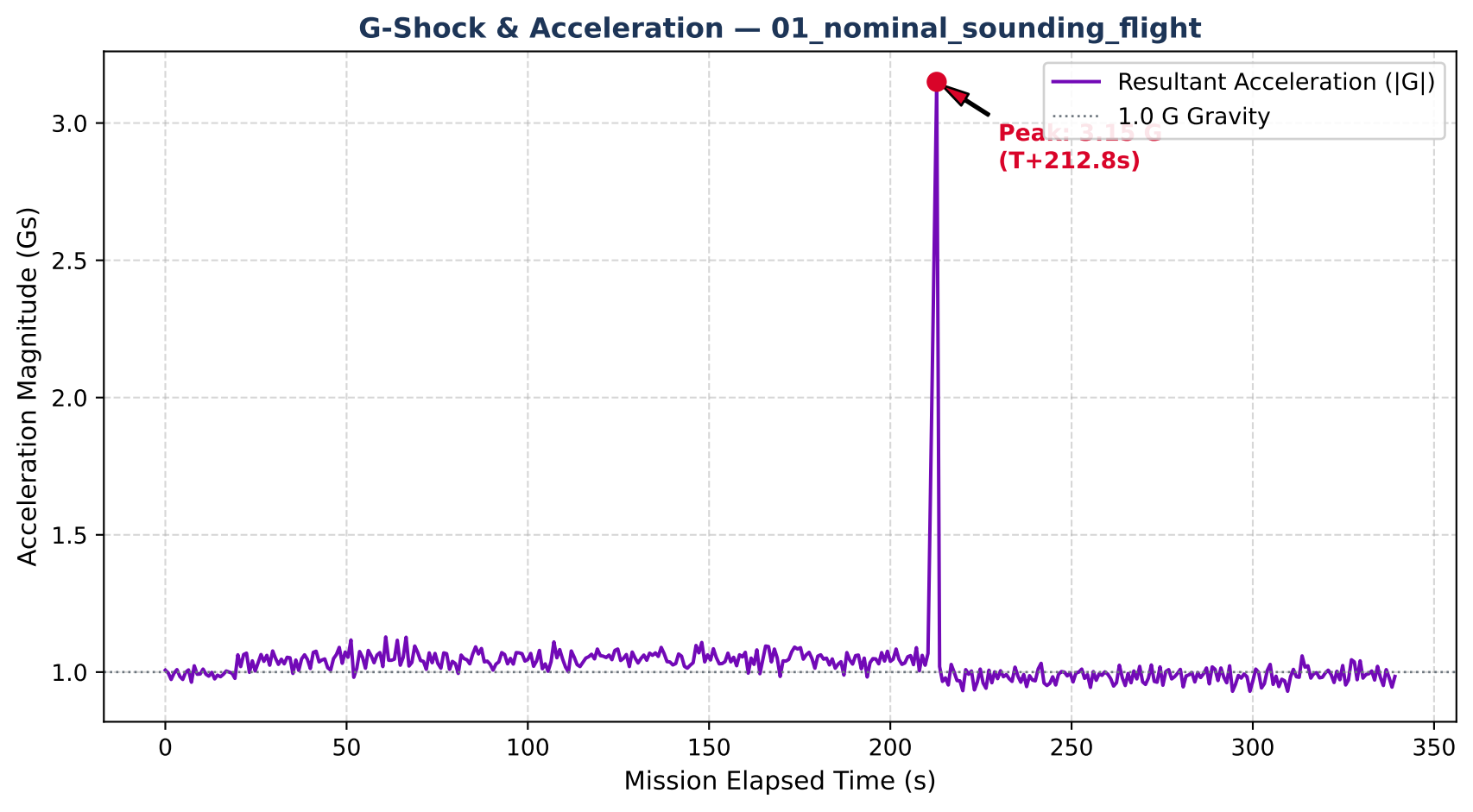
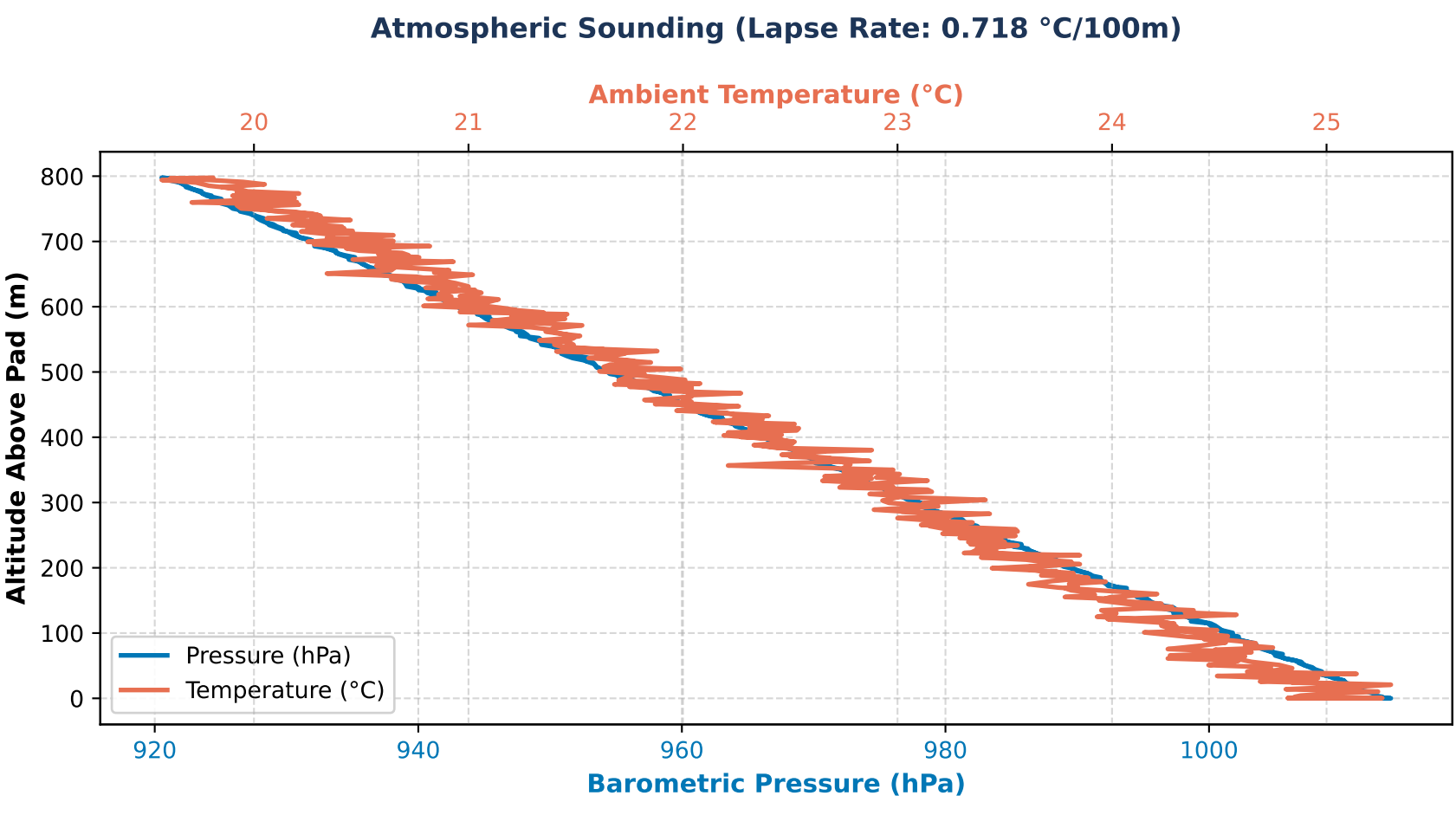
Cognitive CanSat - Mission Summary: 01_nominal_sounding_flight

Flight Dynamics Parameter	Calibrated Value
Mission Name	01_nominal_sounding_flight
Apogee (m)	797.55
Time to Apogee (s)	210.4
Max Ascent Velocity (m/s)	4.82
Max Descent Velocity (m/s)	-6.41
Max G-Shock (Gs)	3.15
Touchdown Shock (Gs)	1.06
Mission Duration (s)	339.21
Min Temperature (°C)	19.57
Max Temperature (°C)	25.3
Min Pressure (hPa)	920.53
Max Pressure (hPa)	1013.8
Lapse Rate (°C/100m)	0.718
Telemetry Frames	423
Landing Lat	22.576126
Landing Lon	88.37007


Altitude vs. Time — 01_nominal_sounding_flight
Raw Barometric
Smoothed
Apogee (797.5 m)
Altitude Above Launch Pad (m)
Mission Elapsed Time (s)
0
100
200
300
400
500
600
700
800
0
50
100
150
200
250
300
350


Vertical Velocity Profile — 01_nominal_sounding_flight
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
4
2
0
−2
−4
−6
0
50
100
150
200
250
300
350
Mission Elapsed Time (s)


G-Shock & Acceleration — 01_nominal_sounding_flight
Resultant Acceleration (|G|)
1.0 G Gravity
Peak: 3.15 G
(T+212.8s)
Acceleration Magnitude (Gs)
3.0
2.5
2.0
1.5
1.0
0
50
100
150
200
250
300
350
Mission Elapsed Time (s)


Atmospheric Sounding (Lapse Rate: 0.718 °C/100m)
Ambient Temperature (°C)
20
21
22
23
24
25
Altitude Above Pad (m)
800
700
600
500
400
300
200
100
0
Pressure (hPa)
Temperature (°C)
920
940
960
980
1000
Barometric Pressure (hPa)

Cognitive CanSat Mission Dynamics: 01_nominal_sounding_flight

